# Point

*From Number Theory to Algebraic Geometry and Logic*

**Session P2 — Topology and Metric Spaces (Merged Edition)**

*Complete lecture notes with proofs, examples, and exercises*

**Abstract**

This session builds the language of closeness in its two required dialects. The *topological* dialect: spaces, bases, and the three constructions (subspace, product, **quotient**) that generate every space of the course; continuity and homeomorphism; connectedness and path-connectedness (the bridge to $\pi_1$ in P3); compactness with the full Heine–Borel picture on $\mathbb{R}^n$. The *metric* dialect: balls, Cauchy sequences, completeness, and the two machines the entire course runs on — **compactness** (open covers to finite data: adelic finiteness in S22, profinite compactness in S34, $\mathbb{Z}_p$ in S19) and **completion** (adding exactly the points Cauchy sequences deserved: $\mathbb{Q} \to \mathbb{R}$ here, $\mathbb{Q} \to \mathbb{Q}_p$ in S17–S18, $k((t))$ in S21). All main proofs are complete; the Zariski contrast (S1) and the adele preview (S22) are woven in throughout.

## Session plan (150 minutes)

| Time | Block |
|---|---|
| 0:00–0:10 | Orientation: closeness without distance; where P2 returns (S1, S17–S22, S26, S34, S38, P3) |
| 0:10–0:40 | §1 Spaces, bases, subspace/product/**quotient**; continuity; Hausdorff and its consequences |
| 0:40–1:05 | §2 Metric spaces: balls as a basis; Cauchy; completeness; $\varepsilon$–$\delta$ $\iff$ open sets |
| 1:05–1:15 | *Break* |
| 1:15–1:35 | §3 Completion: construction, uniqueness; $\mathbb{Q} \to \mathbb{R}$; preview $\mathbb{Q} \to \mathbb{Q}_p$, Ostrowski |
| 1:35–2:00 | §4 Compactness: unit interval (bisection), Heine–Borel on $\mathbb{R}^n$, tube lemma; metric compactness; uniform continuity |
| 2:00–2:20 | §5 Connectedness: intervals, IVT, path-connectedness, components; preview of $\pi_1$ (P3) |
| 2:20–2:30 | Wrap-up: completion = adding missing points; compactness = finiteness engine |

## Conventions

All spaces are nonempty unless stated. "Open" always means open in the topology under discussion. In metric contexts $d$ denotes the metric and $B(x, r) = \{y : d(x, y) < r\}$ the open ball. Sequences are written $(x_n)_{n\geq 1}$ or $(x_n)$. Rings are commutative with 1 when Spec appears (S1).

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# 1 Topological Spaces, Constructions, and Continuity

**Definition 1.1** (Topology)**.** A **topology** on a set $X$ is a collection $\tau \subseteq \mathcal{P}(X)$ of **open sets** with: (T1) $\varnothing, X \in \tau$; (T2) arbitrary unions of members of $\tau$ lie in $\tau$; (T3) finite intersections of members of $\tau$ lie in $\tau$. A set is **closed** if its complement is open. The pair $(X, \tau)$ is a **topological space**.

**Example 1.2** (First examples)**.** • *Discrete:* $\tau = \mathcal{P}(X)$; every subset open (and closed). *Indiscrete:* $\tau = \{\varnothing, X\}$.

- *Standard on $\mathbb{R}^n$:* generated by open balls $B_\varepsilon(x)$ (made precise in §2).
- *Cofinite:* $\tau = \{\varnothing\} \cup \{U : X \setminus U \text{ finite}\}$ on infinite $X$.
- *Zariski:* on $\operatorname{Spec} R$ (S1), closed sets $V(I) = \{\mathfrak{p} \in \operatorname{Spec} R : I \subseteq \mathfrak{p}\}$; not Hausdorff — Remark 1.17.

**Definition 1.3** (Basis)**.** A **basis** for a topology on $X$ is $\mathcal{B} \subseteq \mathcal{P}(X)$ with (B1) $\bigcup_{B \in \mathcal{B}} B = X$; (B2) if $x \in B_1 \cap B_2$ with $B_i \in \mathcal{B}$, then $\exists B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$.

**Proposition 1.4** (A basis generates a topology)**.** *If $\mathcal{B}$ satisfies (B1),(B2), then $\tau = \{U \subseteq X : \forall x \in U\ \exists B \in \mathcal{B},\ x \in B \subseteq U\}$ is a topology, and $\tau$ is exactly the set of unions of members of $\mathcal{B}$.*

*Proof.* $\varnothing \in \tau$ vacuously; $X \in \tau$ by (B1). Unions: a witness for $U_j$ witnesses $\bigcup_i U_i$. Intersections: witnesses $B_1, B_2$ for $x \in U_1 \cap U_2$; (B2) gives $B_3$ with $x \in B_3 \subseteq U_1 \cap U_2$. Every union of basis elements is in $\tau$; conversely $U = \bigcup\{B \in \mathcal{B} : B \subseteq U\}$. □

**Definition 1.5** (Subspace topology)**.** For $Y \subseteq X$, opens of the **subspace topology** are exactly $\{U \cap Y : U \in \tau_X\}$.

**Example 1.6.** In $Y = (0, 1] \subseteq \mathbb{R}$ the set $(\frac{1}{2}, 1]$ is open (it is $(\frac{1}{2}, 2) \cap Y$) though not open in $\mathbb{R}$. In $Y = \mathbb{Z} \subseteq \mathbb{R}$ every subset is open, since $\{n\} = (n - \frac{1}{2}, n + \frac{1}{2}) \cap \mathbb{Z}$.

**Definition 1.7** (Product topology)**.** On $X \times Y$, basis $\{U \times V : U \in \tau_X,\ V \in \tau_Y\}$. For a family $(X_i)_{i \in I}$, basis $\prod_i U_i$ with $U_i$ open and $U_i = X_i$ for all but finitely many $i$.

**Remark 1.8** (Infinite products, and a preview of adeles)**.** The "all but finitely many" restriction defines the product topology; the **box topology** drops it and is too fine (products of compact spaces fail to be compact, etc.). The same discipline returns in S22: the adele ring $A_{\mathbb{Q}}$ is a *restricted product*, $\{(x_v)_v : x_p \in \mathbb{Z}_p \text{ for all but finitely many } p\}$ — controlled infinity, now arithmetic.

**Definition 1.9** (Quotient topology)**.** For a surjection $q : X \to Y$, the **quotient topology** on $Y$ is $\tau_Y = \{V \subseteq Y : q^{-1}(V) \in \tau_X\}$: the finest topology making $q$ continuous.

**Example 1.10** (Quotients everywhere)**.** • $[0, 1]/\{0 \sim 1\} \cong S^1$: identify endpoints of an interval.

- $\mathbb{R}/\mathbb{Z} \cong S^1$ via $[x] \mapsto e^{2\pi i x}$.
- $D^2/\partial D^2 \cong S^2$: collapse the boundary of a disk to a point.
- $\operatorname{Spec}(R/I)$ inside $\operatorname{Spec} R$ (S1, S28): primes containing $I$.

**Definition 1.11** (Continuity; homeomorphism)**.** $f : X \to Y$ is **continuous** if $f^{-1}(V)$ is open for every open $V \subseteq Y$. A **homeomorphism** is a bijective continuous map with continuous inverse; then $X \cong Y$ topologically.

**Proposition 1.12.** *Compositions and identities of continuous maps are continuous; for a bijection $f$, being a homeomorphism $\iff$ $f^{-1}(V)$ open for all opens $V$ $\iff$ $f(U)$ open for all opens $U$.*

**Proposition 1.13** (Compact → Hausdorff bijection is a homeomorphism)**.** *If $X$ is compact, $Y$ is Hausdorff, and $f : X \to Y$ is a continuous bijection, then $f$ is a homeomorphism.*

*Proof.* It suffices that $f$ be closed. Let $C \subseteq X$ be closed; then $C$ is compact (Proposition 4.5); $f(C)$ is compact (Proposition 4.3); and $f(C)$ is closed in $Y$ (Proposition 4.6). Hence $f$ is closed. □

**Example 1.14.** tan gives $(0,1) \cong \mathbb{R}$ (rescaled). The map $[0,1) \to S^1$, $t \mapsto e^{2\pi i t}$, is continuous and bijective but *not* a homeomorphism (Corollary 4.10 explains why cleanly).

**Definition 1.15** (Hausdorff)**.** $X$ is **Hausdorff** (T2) if distinct points have disjoint open neighbourhoods.

**Proposition 1.16** (Hausdorff consequences)**.** *(a) Limits of sequences are unique in Hausdorff spaces. (b) Compact subsets of Hausdorff spaces are closed.*

*Proof.* (a) Disjoint $U \ni x$, $V \ni y$ force $x_n \in U \cap V = \varnothing$ for large $n$. (b) For $x \notin K$, each $y \in K$ has disjoint $U_y \ni x$, $V_y \ni y$; finitely many $V_{y_i}$ cover $K$; then $U = \bigcap_i U_{y_i}$ is an open neighbourhood of $x$ disjoint from $K$. □

**Remark 1.17** (Zariski is not Hausdorff)**.** In $\operatorname{Spec}\mathbb{Z}$ (S1) any two nonempty opens meet: $D(m) \cap D(n) = D(mn) \neq \varnothing$. The Zariski topology sacrifices Hausdorffness to remember algebraic specialization; metric and étale topologies (S17–, S52) restore separation where needed.

## 2 Metric Spaces: Balls, Cauchy, Completeness

**Definition 2.1** (Metric)**.** A **metric** on $X$ is $d : X \times X \to \mathbb{R}_{\geq 0}$ with $d(x,y) = 0 \iff x = y$; symmetry; and the triangle inequality $d(x,z) \leq d(x,y) + d(y,z)$.

**Example 2.2.**

- Euclidean $d_2$, taxicab $d_1$, max $d_\infty$ on $\mathbb{R}^n$ (all give the standard topology).
- Discrete metric $d(x,y) = 1$ for $x \neq y$; induces the discrete topology.
- Sup metric on $C[0,1]$: $d_{\sup}(f,g) = \sup_t |f(t) - g(t)|$; convergence = uniform convergence.
- *Preview (S17):* $p$-adic metric $d_p(x,y) = p^{-v_p(x-y)}$ on $\mathbb{Q}$, with the ultrametric inequality $d_p(x,z) \leq \max\{d_p(x,y), d_p(y,z)\}$ — a geometry where all triangles are isosceles (S19).

**Proposition 2.3** (Balls form a basis)**.** *The open balls $B(x,r)$ satisfy (B1),(B2); hence every metric space carries a canonical metric topology.*

*Proof.* (B1): $x \in B(x,1)$. (B2): if $z \in B(x,r) \cap B(y,s)$ set $\varepsilon = \min\{r - d(x,z),\ s - d(y,z)\} > 0$; then for $w \in B(z,\varepsilon)$: $d(x,w) \leq d(x,z) + d(z,w) < r$, so $B(z,\varepsilon) \subseteq B(x,r)$; symmetrically for $B(y,s)$. □

**Proposition 2.4.** *Every metric space is Hausdorff; hence limits of sequences are unique.*

*Proof.* For $x \neq y$, $r = d(x,y)/2$ gives disjoint $B(x,r), B(y,r)$ by the triangle inequality; apply Proposition 1.16(a). □

**Definition 2.5** (Convergence; Cauchy; complete)**.** $x_n \to x$ means $\forall \varepsilon > 0\ \exists N\ \forall n \geq N :\ d(x_n, x) < \varepsilon$. $(x_n)$ is **Cauchy** if $\forall \varepsilon > 0\ \exists N\ \forall m, n \geq N :\ d(x_m, x_n) < \varepsilon$. $(X,d)$ is **complete** if every Cauchy sequence converges in $X$.

**Proposition 2.6.** *Every convergent sequence is Cauchy.*

*Proof.* Given $\varepsilon$, choose $N$ with $d(x_n, x) < \varepsilon/2$ for $n \geq N$; then $d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \varepsilon$. □

**Example 2.7** (Incompleteness as missing points)**.** In $\mathbb{Q}$: the decimals 1, 1.4, 1.41, 1.414, ... of $\sqrt{2}$ are Cauchy (they converge in $\mathbb{R}$; Proposition 2.6) but have no limit in $\mathbb{Q}$. In $(0,1)$: $x_n = 1/n$ is Cauchy with no limit in the space. In both cases a point is missing (Figure 1).

**Theorem 2.8** (Completeness of the sup metric)**.** $(C[0,1], d_{\sup})$ *is complete: every Cauchy sequence of continuous functions converges uniformly to a continuous function.*

*Proof.* Let $(f_n)$ be Cauchy. For each $t$, $(f_n(t))$ is Cauchy in $\mathbb{R}$ (since $|f_m(t)-f_n(t)| \le d_{\sup}(f_m, f_n)$), hence converges; define $f(t) = \lim_n f_n(t)$. *Uniformity:* given $\varepsilon$, choose $N$ with $d_{\sup}(f_m, f_n) < \varepsilon$ for $m, n \ge N$; fixing $n \ge N$ and letting $m \to \infty$ gives $|f(t)-f_n(t)| \le \varepsilon$ for all $t$: $f_n \to f$ uniformly. *Continuity of $f$:* for $t_0$ and $\varepsilon$, pick $n$ with $d_{\sup}(f, f_n) < \varepsilon/3$, then $\delta$ with $|f_n(t) - f_n(t_0)| < \varepsilon/3$ for $|t - t_0| < \delta$; then $|f(t) - f(t_0)| \le |f(t) - f_n(t)| + |f_n(t) - f_n(t_0)| + |f_n(t_0) - f(t_0)| < \varepsilon$. □

**Proposition 2.9** (Complete $\iff$ closed inside a complete space)**.** *In a complete metric space, a subspace is complete iff it is closed.*

*Proof.* If $Y$ complete and $x_n \to x$ with $x_n \in Y$, then $(x_n)$ Cauchy in $Y$, so converges in $Y$ to some $y \in Y$; uniqueness of limits gives $x = y \in Y$: $Y$ closed. Conversely if $Y$ closed and $(x_n)$ Cauchy in $Y$, it converges in $X$ to some $x$; closedness forces $x \in Y$. □

**Theorem 2.10** ($\varepsilon$–$\delta$ versus open sets)**.** *For $f : X \to Y$ between metric spaces, the following are equivalent: (a) $f$ is continuous in the open-set sense; (b) $\forall x\ \forall \varepsilon > 0\ \exists \delta > 0 :\ d_X(x', x) < \delta \Rightarrow d_Y(f(x'), f(x)) < \varepsilon$.*

*Proof.* (a)⇒(b): $B(f(x), \varepsilon)$ is open, so $f^{-1}(B(f(x), \varepsilon))$ is open and contains $x$; hence it contains some $B(x, \delta)$. (b)⇒(a): let $V$ open and $x \in f^{-1}(V)$; then $B(f(x), \varepsilon) \subseteq V$ for some $\varepsilon$; (b) gives $\delta$ with $f(B(x, \delta)) \subseteq B(f(x), \varepsilon) \subseteq V$; thus each point of $f^{-1}(V)$ has a ball inside $f^{-1}(V)$: open by Proposition 1.4. □

# 3 Completion: Adding the Missing Points

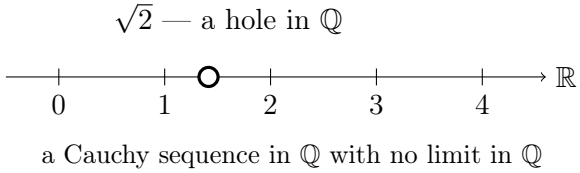



Figure 1: Incompleteness as a missing point: completion (§3) fills all such holes at once.

**Definition 3.1** (Completion)**.** A **completion** of $(X, d)$ is a complete metric space $(\hat{X}, \hat{d})$ with an isometric embedding $\iota : X \to \hat{X}$ whose image is dense in $\hat{X}$.

**Theorem 3.2** (Existence of completions)**.** *Every metric space has a completion.*

*Proof.* Let $\mathcal{C}$ be the set of Cauchy sequences in $X$; declare $(x_n) \sim (y_n)$ iff $d(x_n, y_n) \to 0$ (an equivalence relation — P0). Set $\hat{X} = \mathcal{C}/\!\sim$ and $\hat{d}([(x_n)], [(y_n)]) = \lim_n d(x_n, y_n)$. *The limit exists:* $|d(x_m, y_m) - d(x_n, y_n)| \le d(x_m, x_n) + d(y_m, y_n) \to 0$, so $(d(x_n, y_n))$ is Cauchy in $\mathbb{R}$, hence convergent. *Independent of representatives:* if $(x_n') \sim (x_n)$ and $(y_n') \sim (y_n)$ then $|d(x_n', y_n') - d(x_n, y_n)| \le d(x_n', x_n) + d(y_n', y_n) \to 0$. *Metric axioms:* symmetry and triangle pass to the limit; $\hat{d} = 0 \iff (x_n) \sim (y_n)$. *Embedding:* $\iota(x) = [(x, x, \dots)]$ satisfies $\hat{d}(\iota(x), \iota(y)) = d(x, y)$. *Density:* for $[(x_n)]$ and $\varepsilon$, choose $N$ with $d(x_m, x_n) < \varepsilon$ for $m, n \ge N$; then $\hat{d}(\iota(x_N), [(x_n)]) = \lim_m d(x_N, x_m) \le \varepsilon$. *Completeness:* let $([\xi^{(k)}])_k$ be Cauchy in $\hat{X}$; by density pick $x_k \in X$ with $\hat{d}(\iota(x_k), \xi^{(k)}) < 1/k$; then $(x_k)$ is Cauchy in $X$ (triangle inequality), and $\hat{d}(\iota(x_k), \xi^{(k)}) \to 0$ forces $\iota(x_k) \to$ the given Cauchy class; hence $[(x_k)] \in \hat{X}$ is the limit. □

**Theorem 3.3** (Uniqueness of completions)**.** *Any two completions* $(\hat{X}_1, \iota_1)$, $(\hat{X}_2, \iota_2)$ *of* $(X, d)$ *are related by a unique isometry* $\Phi : \hat{X}_1 \to \hat{X}_2$ *with* $\Phi \circ \iota_1 = \iota_2$.

*Proof idea.* $\iota_2 \circ \iota_1^{-1} : \iota_1(X) \to \iota_2(X)$ is an isometry between dense subspaces; isometries are uniformly continuous, and uniformly continuous maps from a dense subspace into a complete space extend uniquely continuously; the extensions of $\Phi$ and of its inverse are mutually inverse isometries by uniqueness of continuous extensions from dense subspaces. □

**Example 3.4** (Completions we will live with)**.**

- The completion of $(\mathbb{Q}, |\cdot|)$ is $\mathbb{R}$ (Cantor's construction is exactly Theorem 3.2).
- The completion of $((0,1), |\cdot|)$ is $[0,1]$.
- The completion of $(\mathbb{Q}, d_p)$ is $\mathbb{Q}_p$, and $\hat{\mathbb{Z}} = \varprojlim \mathbb{Z}/p^n$ appears in S17–S18: the same machine, a different metric, a different world.
- $(C[0,1], d_{\sup})$ is already complete (Theorem 2.8); its $L^2$-metric completion is $L^2[0,1]$ (mention; not used later).

**Remark 3.5** (Ostrowski preview: the census of completions of $\mathbb{Q}$)**.** Ostrowski's theorem (S17) says that, up to equivalence, the only non-trivial metrics on $\mathbb{Q}$ are $|\cdot|$ and the $d_p$; hence the completions of $\mathbb{Q}$ are exactly $\mathbb{R}$ and the $\mathbb{Q}_p$. The course's slogan: each completion adds precisely the points its Cauchy sequences deserved; the adeles (S22) then gather all completions into one space.

# 4 Compactness: General and Metric

**Definition 4.1** (Compact)**.** $X$ is **compact** if every open cover $\{U_i\}_{i \in I}$ has a finite subcover.

**Example 4.2.**

- Finite spaces are compact; $\{1/n : n \geq 1\} \cup \{0\} \subseteq \mathbb{R}$ is compact (one open catches 0 and hence all but finitely many points).
- $(0,1)$ is not compact: $(1/n, 1)_{n \geq 2}$ covers with no finite subcover; $\mathbb{R}$ is not compact: $(-n, n)_{n \geq 1}$.
- $S^1$ is compact: continuous image of $[0,1]$ under $t \mapsto e^{2\pi i t}$.

**Proposition 4.3** (Continuous images of compact sets)**.** *The continuous image of a compact space is compact.*

*Proof.* If $\{V_i\}$ covers $f(X)$ then $\{f^{-1}(V_i)\}$ covers $X$; a finite subcover of the latter maps to a finite subcover of the former. □

**Theorem 4.4** (Compactness of the unit interval)**.** $[0,1]$ *with the standard topology is compact.*

*Proof by bisection.* Assume an open cover $\mathcal{U}$ with no finite subcover. Bisect: of the two halves of $[0,1]$, at least one cannot be finitely covered; call it $I_1$. Bisect $I_1$, choose a non-finitely-coverable half $I_2$; continue, obtaining nested closed intervals $I_n$ with $|I_n| = 2^{-n}$, none finitely covered. By the nested-interval property, $\bigcap_n I_n = \{s\}$ for a single $s \in [0,1]$. Some $U \in \mathcal{U}$ contains $s$, and $U$ open gives $\varepsilon > 0$ with $(s - \varepsilon, s + \varepsilon) \cap [0,1] \subseteq U$. Choose $n$ with $2^{-n} < \varepsilon$; since $s \in I_n$ and $|I_n| < \varepsilon$, $I_n \subseteq (s - \varepsilon, s + \varepsilon)$, so $\{U\}$ finitely covers $I_n$ — contradiction. □

**Proposition 4.5** (Closed inside compact)**.** *A closed subset of a compact space is compact.*

*Proof.* If $K \subseteq X$ closed and $\{U_i\}$ covers $K$, then $\{U_i\} \cup \{X \setminus K\}$ covers $X$; a finite subcover minus $X \setminus K$ finitely covers $K$. □

**Proposition 4.6** (Compact inside Hausdorff)**.** *Compact subsets of Hausdorff spaces are closed.*

*Proof.* Proposition 1.16(b). □

**Lemma 4.7** (Tube lemma)**.** *If $Y$ is compact and $W \subseteq X \times Y$ is open with $\{x\} \times Y \subseteq W$, then there exists an open $U \ni x$ with $U \times Y \subseteq W$.*

*Proof.* For each $y \in Y$, $(x, y) \in W$, so basic opens $U_y \ni x$, $V_y \ni y$ exist with $U_y \times V_y \subseteq W$. The $\{V_y\}$ cover $Y$; compactness gives $V_{y_1}, \ldots, V_{y_k}$ covering $Y$. Let $U = U_{y_1} \cap \cdots \cap U_{y_k}$; then for any $(u, v) \in U \times Y$: $v \in V_{y_j}$ for some $j$, so $(u, v) \in U_{y_j} \times V_{y_j} \subseteq W$. □

**Proposition 4.8** (Tychonoff, finite case)**.** *If $X$ and $Y$ are compact, then $X \times Y$ is compact.*

*Proof.* Let $\{W_i\}$ cover $X \times Y$. For each $x \in X$, the slice $\{x\} \times Y \cong Y$ is compact, so finitely many $W_{i_1}, \ldots, W_{i_k}$ cover it; by the tube lemma there is an open $U_x \ni x$ with $U_x \times Y \subseteq W_{i_1} \cup \cdots \cup W_{i_k}$. The $\{U_x\}$ cover $X$; compactness gives $U_{x_1}, \ldots, U_{x_m}$ covering $X$; the corresponding finite family of $W_i$'s covers $X \times Y$. □

**Theorem 4.9** (Heine–Borel)**.** *$K \subseteq \mathbb{R}^n$ is compact iff $K$ is closed and bounded.*

*Proof.* ($\Leftarrow$): $[a, b] \cong [0, 1]$ (affine map), hence compact (Theorem 4.4); boxes $[a_1, b_1] \times \cdots \times [a_n, b_n]$ are compact (Proposition 4.8, inductively); a closed subset of a compact is compact (Proposition 4.5), and bounded $K$ lies in some box. ($\Rightarrow$): unbounded $K$ is covered by $\{B_n(0)\}$ with no finite subcover; if $K$ is not closed, pick $x \in \bar{K} \setminus K$; then $\{X \setminus \bar{B}_{1/n}(x)\}$ covers $K$ with no finite subcover. □

**Corollary 4.10** (Compactness is a homeomorphism invariant)**.** *Hence $[0, 1] \ncong (0, 1) \ncong \mathbb{R}$, and the bijection $[0, 1) \to S^1$ of Example 1.14 is not a homeomorphism.*

**Corollary 4.11** (Extreme values)**.** *A continuous $f : X \to \mathbb{R}$ on a nonempty compact space attains its maximum and minimum.*

*Proof.* $f(X)$ is compact (Proposition 4.3), hence closed and bounded (Theorem 4.9); so $\sup f(X), \inf f(X) \in f(X)$. □

**Definition 4.12** (Sequential compactness; total boundedness)**.** $X$ is **sequentially compact** if every sequence has a convergent subsequence; **totally bounded** if for every $\varepsilon > 0$ finitely many $\varepsilon$-balls cover $X$.

**Proposition 4.13** (The three notions coincide for metric spaces)**.** *For metric spaces: compact $\Rightarrow$ sequentially compact $\Rightarrow$ complete and totally bounded; and complete + totally bounded $\Rightarrow$ compact.*

*Proof ideas.* Compact $\Rightarrow$ seq. compact: if some sequence had no convergent subsequence, each $x$ would have a neighbourhood $U_x$ meeting the sequence finitely often; finitely many $U_x$ cover $X$, yet the sequence has infinitely many terms — contradiction. Seq. compact $\Rightarrow$ totally bounded: else some $\varepsilon > 0$ admits an infinite $\varepsilon$-separated sequence, which has no Cauchy (hence no convergent) subsequence. Seq. compact $\Rightarrow$ complete: a Cauchy sequence with a convergent subsequence converges to the same limit. Complete + totally bounded $\Rightarrow$ compact: from any open cover build, via total boundedness and a Lebesgue-number argument, a finite subcover (details: Munkres §32). □

**Theorem 4.14** (Uniform continuity on compact metric spaces)**.** *A continuous map from a compact metric space to a metric space is uniformly continuous.*

*Proof.* Given $\varepsilon$, for each $x$ choose $\delta_x$ with $f(B(x, \delta_x)) \subseteq B(f(x), \varepsilon/2)$. The balls $B(x, \delta_x/2)$ cover $X$; compactness gives finitely many $B(x_i, \delta_{x_i}/2)$; set $\delta = \min_i \delta_{x_i}/2 > 0$. If $d(x, x') < \delta$, pick $i$ with $x \in B(x_i, \delta_{x_i}/2)$; then $d(x', x_i) \le d(x', x) + d(x, x_i) < \delta_{x_i}$; hence both $f(x), f(x')$ lie in $B(f(x_i), \varepsilon/2)$, so $d_Y(f(x), f(x')) < \varepsilon$. □

# 5 Connectedness and Path-Connectedness

**Definition 5.1** (Connected)**.** $X$ is **connected** if it cannot be written $X = U \sqcup V$ with $U, V$ nonempty open; equivalently, the only clopen subsets are $\varnothing$ and $X$.

**Proposition 5.2** (Continuous image of connected is connected)**.** *If $X$ is connected and $f : X \to Y$ continuous, then $f(X)$ is connected.*

*Proof.* If $f(X) = U \sqcup V$ is a disconnection then $X = f^{-1}(U) \sqcup f^{-1}(V)$ is a disconnection of $X$. □

**Theorem 5.3** (Intervals are connected)**.** *Every interval $I \subseteq \mathbb{R}$ is connected; in particular $\mathbb{R}$ and $[0, 1]$ are connected.*

*Proof.* Suppose $I = U \sqcup V$ with $U, V$ nonempty open in $I$. Pick $a \in U$, $b \in V$; wlog $a < b$; replace $I$ by $[a, b] \subseteq I$ and $U, V$ by their intersections with $[a, b]$. Let $s = \sup(U \cap [a, b])$; then $s \in [a, b] = U \sqcup V$. If $s \in U$: $U$ open gives $\varepsilon > 0$ with $[s, s+\varepsilon) \subseteq U$ when $s < b$; then $s + \varepsilon/2 \in U$ exceeds the supremum — impossible; and $s = b$ forces $b \in U \cap V$. If $s \in V$: $V$ open gives $\varepsilon > 0$ with $(s-\varepsilon, s] \subseteq V$; but $s$ being the least upper bound of $U \cap [a, b]$ yields $u \in U \cap (s-\varepsilon, s] \subseteq U \cap V$ — impossible. □

**Corollary 5.4** (Intermediate value theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) < c < f(b)$ (or $f(b) < c < f(a)$), then some $x \in (a, b)$ has $f(x) = c$.*

*Proof.* $f([a, b])$ is connected (Proposition 5.2); a connected subset of $\mathbb{R}$ is an interval (Theorem 5.3 applied to its complement's disconnection otherwise); hence it contains $c$. □

**Example 5.5.** $\mathbb{Q}$ is totally disconnected: for $x < y$ in $\mathbb{Q}$ pick an irrational $t \in (x, y)$; then $(-\infty, t) \cap \mathbb{Q}$ and $(t, \infty) \cap \mathbb{Q}$ disconnect any subset containing $x, y$. Contrast: $\mathbb{R}_{>0}$, circles, and all intervals are connected.

**Definition 5.6** (Path-connected; components)**.** $X$ is **path-connected** if for every $x, y \in X$ there is a continuous $\gamma : [0, 1] \to X$ with $\gamma(0) = x$, $\gamma(1) = y$. The **connected components** of $X$ are the maximal connected subsets; they are closed, pairwise disjoint, and cover $X$.

**Example 5.7** (Path-connected vs connected)**.**

- $\mathbb{R}^n$, open balls, convex sets: path-connected (straight lines); $S^1$: path-connected ($t \mapsto e^{2\pi i t}$).
- The topologist's sine curve $\{(x, \sin(1/x)) : x > 0\} \cup \{(0, y) : -1 \le y \le 1\}$: connected but not path-connected.

**Remark 5.8** (Preview: $\pi_1$ and homology (P3))**.** Connectedness is a coarse invariant ("how many pieces"); the fundamental group $\pi_1(X, x_0)$ (loops up to homotopy) is finer: $\pi_1(S^1) \cong \mathbb{Z}$, $\pi_1(\mathbb{R}^n) = 0$; homology $H_n(X)$ generalizes further. These algebraic invariants turn topology into algebra — the theme of P3, and later of $\pi_1^{\text{ét}}$ (S35).

# 6 Exercises

**Exercise 1.** Verify that the cofinite collection on an infinite set satisfies (T1)–(T3), and that it is not Hausdorff. Which sequences converge, and to how many limits?

**Exercise 2.** Show $\{(a, b] : a < b\}$ is a basis for a topology on $\mathbb{R}$ (the lower-limit topology); prove it is strictly finer than the standard topology and Hausdorff, and that $[0, 1)$ is open in it.

**Exercise 3.** Prove that the subspace topology on $A \subseteq X$ is the coarsest topology making the inclusion continuous, and that the quotient topology on $Y$ is the finest making $q : X \to Y$ continuous.

**Exercise 4.** Show $\{1/n : n \geq 1\} \cup \{0\}$ is compact directly from the definition, and that $\{1/n : n \geq 1\}$ (without 0) is not.

**Exercise 5.** Prove Proposition 2.3 in full detail, and show $d_1, d_2, d_\infty$ on $\mathbb{R}^n$ all generate the standard topology (compare balls: $B_\infty \subseteq B_2 \subseteq B_1 \subseteq n\,B_\infty$).

**Exercise 6.** Prove Proposition 2.9; deduce that $\mathbb{Z} \subseteq \mathbb{R}$ is complete, and that $\mathbb{Z}_p \subseteq \mathbb{Q}_p$ is closed (the latter using the ultrametric inequality — preview of S19).

**Exercise 7.** In $(\mathbb{Q}, |\cdot|)$, prove directly (without invoking $\mathbb{R}$) that $x_1 = 1$, $x_{n+1} = \frac{1}{2}(x_n + 2/x_n)$ is Cauchy and has no limit in $\mathbb{Q}$.

**Exercise 8.** Carry out Theorem 3.2 for $X = (0, 1)$ explicitly: identify $\hat{X}$ with $[0, 1]$ by describing the equivalence classes of Cauchy sequences corresponding to 0 and 1.

**Exercise 9.** Show the product topology on $\mathbb{R}^{\mathbb{N}}$ (countable product) is strictly coarser than the box topology, by exhibiting a box-open set containing no product-basic neighbourhood of the origin (cf. Remark 1.8).

**Exercise 10.** Prove the tube lemma (Lemma 4.7) and use it to prove Tychonoff for finite products (Proposition 4.8).

**Exercise 11.** Prove Theorem 4.14 again by contradiction using sequential compactness, and deduce that every continuous $f : S^1 \to \mathbb{R}$ is uniformly continuous.

**Exercise 12.** Prove Theorem 5.3 by a different method: show the only clopen subsets of $\mathbb{R}$ are $\varnothing$ and $\mathbb{R}$.

**Exercise 13.** Show the topologist's sine curve (Example 5.7) is connected but not path-connected; and prove that path-connected always implies connected.

**Exercise 14.** Show $\mathbb{R}/\mathbb{Z} \cong S^1$ via $[x] \mapsto e^{2\pi i x}$: verify it is a well-defined continuous bijection, and use Proposition 1.13 to conclude it is a homeomorphism.

## Where This Session Feeds the Course

- **Depends on:** P0 (sets, equivalence relations for quotients); P1 (group language in examples and $\pi_1$).
- **Feeds:** S1 (Zariski topology, quasi-compactness of Spec); S17–S18 (completion $\mathbb{Q} \to \mathbb{Q}_p$; ultrametric topology); S19 (compactness of $\mathbb{Z}_p$); S21 ($k((t))$ as completion); S22 (restricted product, adelic compactness — the finiteness engine for S24); S26 (sheaves on topological spaces); S28 (schemes as locally ringed spaces); S33–S34 (quasi-compactness, profinite compactness); S37–S38 (complex analysis, compact Riemann surfaces); S52 (contrast with Zariski/étale); P3 ($\pi_1$, coverings, homology).
- **Unifying thread:** completion adds exactly the points Cauchy sequences deserved — $\mathbb{R}$ here, $\mathbb{Q}_p$ in Act III; compactness is what makes infinite point-counts finite, from the unit interval to $A_\mathbb{Q}/\mathbb{Q}$; quotient topology is identification made continuous — the geometric echo of P0's partitions.

### References for this session


[1] Munkres, J.R. *Topology*, ch.2–4 (bases, subspace/product, connectedness, compactness).

[2] Sutherland, W.A. *Introduction to Metric and Topological Spaces* (metric viewpoint).

[3] Rudin, W. *Principles of Mathematical Analysis*, ch.2–4 (completion; completeness of $C[0,1]$).

[4] Gouvêa, F.Q. *p-adic Numbers: An Introduction*, ch.1–3 (the $p$-adic preview).

[5] Willard, S. *General Topology* (more advanced, comprehensive).

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*End of Session P2 — next: Session P3, Fundamental Group and Covering Spaces*